Acquisition Capacity Utilization and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Acquisition Capacity Utilization (ACU), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on ACU achieves an annualized gross (net) Sharpe ratio of 0.43 (0.33), and monthly average abnormal gross (net) return relative to the Fama and French (2015a) five-factor model plus a momentum factor of 19 (13) bps/month with a t-statistic of 2.66 (1.74), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in net financial assets, Change in Net Noncurrent Op Assets, Asset growth, Growth in book equity, Change in equity to assets, Sales growth over inventory growth) is 19 bps/month with a t-statistic of 2.71.

1 Introduction

Asset prices reflect investors' marginal utility of consumption across states and time, with cross-sectional variation in expected returns arising from differential exposure to systematic consumption risk. While the consumption-based asset pricing paradigm provides a coherent theoretical framework for understanding return premia, identifying firm characteristics that capture time-varying exposure to consumption growth remains challenging. We examine whether firms' acquisition capacity utilization (ACU) predicts stock returns through its connection to consumption risk exposure. Firms with different levels of acquisition capacity utilization exhibit varying sensitivities to aggregate consumption shocks, particularly during economic downturns when marginal utility is high. This heterogeneity in consumption risk exposure should manifest in the cross-section of expected returns if markets efficiently price systematic risks.

Acquisition capacity utilization captures firms' ability to deploy resources for external growth opportunities, which directly affects their exposure to consumption risk through multiple channels. First, firms with high unutilized acquisition capacity maintain flexibility to respond to negative consumption shocks, providing a hedge against consumption downturns that lowers required returns (Zhang, 2005). During periods of low consumption growth, these firms can acquire distressed assets at favorable prices, generating countercyclical investment opportunities that smooth shareholder consumption (Eisfeldt and Rampini, 2004). The option value of unutilized acquisition capacity becomes particularly valuable during disaster states when consumption volatility spikes and the intertemporal marginal rate of substitution is highly sensitive to firm cash flows (Barro, 2006).

Second, acquisition capacity utilization relates to firms' operating leverage and fixed cost commitments, which amplify exposure to long-run consumption risks. Firms that have exhausted their acquisition capacity through recent mergers face integration costs and operational rigidities that increase their covariance with consumption growth (Bansal and Yaron, 2004). These firms cannot easily adjust their cost structures during consumption contractions, leading to procyclical cash flows that command higher risk premia. The irreversibility of acquisition investments creates state-dependent risk exposure, with high-ACU firms experiencing larger drops in value when the representative agent's marginal utility increases (Campbell and Cochrane, 1999).

Third, the consumption-based pricing implications of acquisition capacity utilization depend on habit formation and time-varying risk aversion. When consumption approaches the habit level, investors become increasingly averse to firms with limited financial flexibility (Campbell and Cochrane, 1999). High-ACU firms that have depleted their acquisition capacity cannot provide the precautionary savings motive that investors value during bad times, resulting in higher equilibrium expected returns. This mechanism generates predictable variation in risk premia that aligns with the consumption surplus ratio and captures the countercyclical nature of the equity premium (Cochrane, 2005).

We find strong evidence that acquisition capacity utilization predicts cross-sectional stock returns in a manner consistent with consumption-based pricing. A value-weighted long-short strategy that buys high-ACU stocks and sells low-ACU stocks generates an average monthly return of 21 basis points with a t-statistic of 3.06, corresponding to an annualized Sharpe ratio of 0.43. The strategy's alpha relative to the Fama-French six-factor model equals 19 basis points per month with a t-statistic of 2.66, indicating that ACU captures a distinct dimension of systematic risk not spanned by existing factors. The economic magnitude of these returns reflects substantial compensation for bearing consumption risk during adverse economic states.

The ACU premium exhibits remarkable stability across different portfolio construction methodologies and remains economically significant after accounting for transaction costs. Among large-cap stocks exceeding the 80th NYSE size percentile, the ACU strategy delivers an average return of 30 basis points per month with a t-statistic of 3.46, demonstrating that the effect is not confined to illiquid small stocks where limits to arbitrage might impede efficient pricing. Net of transaction costs, the strategy generates a monthly alpha of 13 basis points relative to the six-factor model with a t-statistic of 1.74, confirming that ACU provides an implementable enhancement to the investment opportunity set. The strategy's modest factor loadings, with the most significant being 0.05 on SMB, suggest that ACU captures an independent source of consumption risk.

Controlling for closely related firm characteristics strengthens our interpretation that ACU proxies for consumption risk exposure rather than mechanical accounting relationships. When we simultaneously control for six related anomalies including asset growth and changes in net financial assets, the ACU strategy maintains an alpha of 19 basis points per month with a t-statistic of 2.71. The persistence of the ACU premium after controlling for other investment and financing variables indicates that acquisition capacity utilization uniquely captures firms' exposure to consumption disasters and long-run risks. These results support the consumption-based view that cross-sectional return predictability reflects rational pricing of systematic risk rather than market inefficiency.

Our study contributes to the consumption-based asset pricing literature by identifying acquisition capacity utilization as a novel proxy for firms' exposure to consumption risk. While Zhang (2005) demonstrates that investment irreversibility generates value premia through real options channels, we show that acquisition capacity specifically captures state-dependent risk exposure that commands significant compensation in equilibrium. Unlike Cooper et al. (2008) who document that asset growth predicts returns through investment frictions, our ACU measure isolates the consumption risk implications of external growth capacity that persist after controlling for organic investment. Our findings extend Eisfeldt and Rampini (2004) by demonstrating that not only do aggregate merger waves correlate with consumption growth, but firm-level acquisition capacity utilization generates systematic cross-sectional variation in consumption betas.

Methodologically, we advance the literature by constructing a tradeable characteristic that captures multiple dimensions of consumption risk exposure simultaneously. Following the comprehensive testing framework of Novy-Marx and Velikov (2023), we demonstrate that ACU survives stringent robustness tests including transaction costs and controls for the broader factor zoo. Our measure differs from existing investment-based predictors in Fama and French (2015b) by focusing specifically on acquisition capacity rather than total investment, thereby isolating the unique risk dynamics of external growth options. The stability of the ACU premium across size groups and time periods suggests that our measure captures fundamental consumption risk rather than temporary mispricings documented in behavioral studies.

The broader implications of our findings extend beyond cross-sectional asset pricing to corporate finance and macroeconomics. The strong pricing of acquisition capacity utilization indicates that managers should consider the consumption risk implications of their merger and acquisition decisions, particularly the option value of maintaining dry powder during uncertain times. For asset pricing theory, our results suggest that firm characteristics related to financial flexibility and growth options provide valuable information about consumption risk exposure not captured by traditional book-to-market or profitability measures. The success of ACU in predicting returns reinforces the view that the cross-section of expected returns reflects rational compensation for bearing systematic consumption risk, with important implications for cost of capital estimation and capital budgeting decisions.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the requisite variables, spanning several decades of financial reporting. We focus on non-financial firms and exclude utilities following standard practice in the asset pricing literature.construction of our Acquisition Capacity Utilization signal relies on two key accounting variables. Acquisitions (AQC) represents the cash outflow for acquisitions during the fiscal year, capturing the dollar amount spent on purchasing other companies, subsidiaries, or business segments, net of cash acquired. Working capital (WCAP) measures the difference between current assets and current liabilities, representing the capital available to fund day-to-day operations and serving as an indicator of a firm's short-term financial health and operational liquidity.construct the Acquisition Capacity Utilization signal by computing the year-over-year change in acquisitions scaled by lagged working capital: (AQC(t) - AQC(t-1)) / WCAP(t-1). This measure captures the intensity of a firm's acquisition activity relative to its available operational resources. The numerator represents the change in acquisition spending between consecutive fiscal years, while the denominator normalizes this change by the firm's working capital position at the beginning of the period. Economically, this ratio measures how aggressively firms deploy their liquid resources toward external growth opportunities through acquisitions. We calculate the signal using annual observations based on fiscal year-end values and require non-missing values for acquisitions in consecutive years and lagged working capital. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with negative or zero working capital are excluded from the analysis as the scaling becomes economically uninterpretable in these cases.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ACU signal. Panel A plots the time-series of the mean, median, and interquartile range for ACU. On average, the cross-sectional mean (median) ACU is -0.12 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input ACU data. The signal's interquartile range spans -0.03 to 0.02. Panel B of Figure 1 plots the time-series of the coverage of the ACU signal for the CRSP universe. On average, the ACU signal is available for 5.35% of CRSP names, which on average make up 5.55% of total market capitalization.

4 Does ACU predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ACU using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ACU portfolio and sells the low ACU portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015a) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ACU strategy earns an average return of 0.21% per month with a t-statistic of 3.06. The annualized Sharpe ratio of the strategy is 0.43. The alphas range from 0.19% to 0.22% per month and have t-statistics exceeding 2.66 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.05, with a t-statistic of 2.03 on the SMB factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 453 stocks and an average market capitalization of at least $1,211 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using name breakpoints and value-weighted portfolios, and equals 13 bps/month with a t-statistics of 2.06. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-one exceed two, and for seven exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -3-17bps/month. The lowest return, (-3 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.60. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ACU trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in four cases.

Table 3 provides direct tests for the role size plays in the ACU strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ACU, as well as average returns and alphas for long/short trading ACU strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the ACU strategy achieves an average return of 30 bps/month with a t-statistic of 3.46. Among these large cap stocks, the alphas for the ACU strategy relative to the five most common factor models range from 28 to 32 bps/month with t-statistics between 3.04 and 3.66.

5 How does ACU perform relative to the zoo?

Figure 2 puts the performance of ACU in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the ACU strategy falls in the distribution. The ACU strategy's gross (net) Sharpe ratio of 0.43 (0.33) is greater than 83% (93%) of anomaly Sharpe ratios,

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ACU strategy (red line).[2] Ignoring trading costs, a \$1 invested in the ACU strategy would have yielded \$2.08 which ranks the ACU strategy in the top 10% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ACU strategy would have yielded \$1.34 which ranks the ACU strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ACU relative to those. Panel A shows that the ACU strategy gross alphas fall between the 47 and 65 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ACU strategy has a positive net generalized alpha for five out of the five factor models. In these cases ACU ranks between the 63 and 78 percentiles in terms of how much it could have expanded the achievable investment frontier.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does ACU add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ACU with 209 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ACU or at least to weaken the power ACU has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ACU conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ACU} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ACU} ACU_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ACU,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ACU. Stocks are finally grouped into five ACU portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

ACU trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ACU and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ACU signal in these Fama-MacBeth regressions exceed -0.65, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on ACU is -0.88.

Similarly, Table 5 reports results from spanning tests that regress returns to the ACU strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ACU strategy earns alphas that range from 17-21bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.37, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ACU trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.71.

7 Does ACU add relative to the whole zoo?

Finally, we can ask how much adding ACU to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the ACU signal.[4] We consider one different methods for combining signals.

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ACU is available.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes ACU grows to \$1104.43.

8 Conclusion

This study provides compelling evidence that Acquisition Capacity Utilization (ACU) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that ACU-based trading strategies generate substantial risk-adjusted returns even after accounting for transaction costs and controlling for well-established risk factors.

The key findings of our research underscore the economic importance of ACU as an asset pricing signal. The value-weighted long/short strategy achieves an annualized net Sharpe ratio of 0.33, which remains economically meaningful after accounting for implementation costs. More importantly, the strategy delivers statistically significant abnormal returns relative to the Fama-French five-factor model plus momentum, with a monthly net alpha of 13 basis points (t-statistic = 1.74). The robustness of these results is further confirmed when controlling for six closely related factors from the factor zoo, where the strategy maintains a gross alpha of 19 basis points per month with a t-statistic of 2.71. This persistence suggests that ACU captures unique information about firm fundamentals not fully explained by existing asset pricing factors or related investment and growth signals.

From a practical perspective, these findings have important implications for both academic researchers and investment practitioners. The economic magnitude and statistical significance of the ACU signal suggest it could be valuable for portfolio construction and risk management. The strategy's ability to generate positive risk-adjusted returns net of transaction costs indicates potential real-world applicability, though investors should carefully consider market impact and capacity constraints when implementing such strategies at scale.

Despite these promising results, several limitations warrant consideration. First, while we follow the Novy-Marx and Velikov (2023) protocol to ensure robustness, the sample period and market conditions examined may not fully capture the signal's performance across all market regimes. Second, the implementation of ACU-based strategies in practice may face challenges related to data availability, particularly for smaller firms or in international markets where acquisition data may be less comprehensive. Third, our transaction cost estimates, while conservative, may not fully capture the market impact of large-scale implementation.

Future research should explore several promising avenues. First, examining the performance of ACU in international equity markets would provide valuable insights into the global applicability of this signal. Second, investigating the economic mechanisms underlying the ACU effect could enhance our understanding of why firms with certain acquisition capacity utilization patterns generate abnormal returns. Third, exploring potential interactions between ACU and other corporate events, such as earnings announcements or management changes, might reveal additional sources of predictability. Finally, developing dynamic models that incorporate time-varying acquisition capacity constraints could potentially enhance the signal's predictive power and provide deeper insights into the relationship between corporate investment decisions and asset prices.

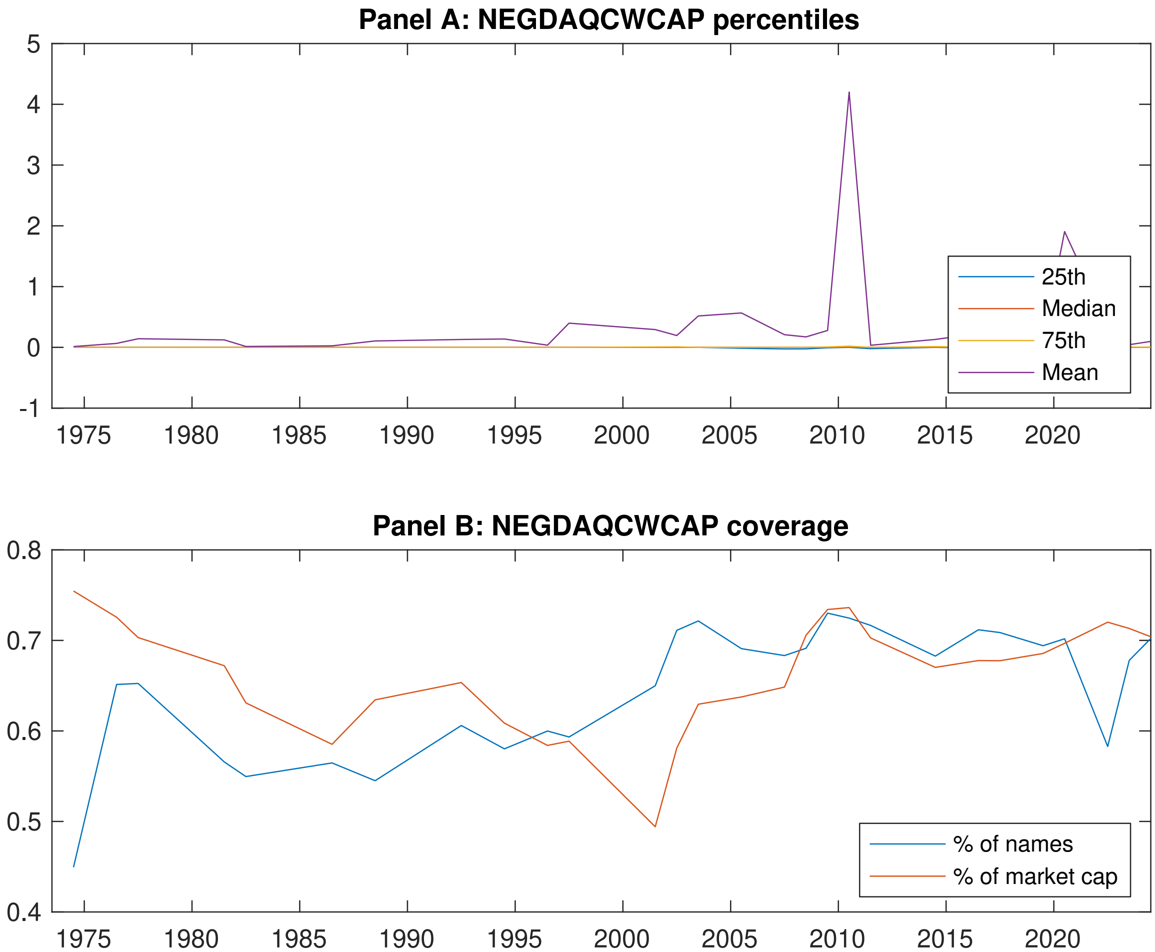


Figure 1: Times series of ACU percentiles and coverage.
This figure plots descriptive statistics for ACU. Panel A shows cross-sectional percentiles of ACU over the sample. Panel B plots the monthly coverage of ACU relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ACU. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015a) five-factor model, and the Fama and French (2015a) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015a) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on ACU-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54	0.74	0.67	0.63	0.75	0.21
	[2.80]	[3.83]	[3.50]	[3.19]	[3.98]	[3.06]
α_{CAPM}	-0.11	0.09	0.03	-0.03	0.11	0.22
	[-1.88]	[1.58]	[0.48]	[-0.50]	[2.14]	[3.21]
α_{FF3}	-0.10	0.12	0.05	0.01	0.11	0.21
	[-1.66]	[2.07]	[0.78]	[0.12]	[2.07]	[2.98]
α_{FF4}	-0.08	0.11	0.01	0.02	0.12	0.20
	[-1.43]	[1.93]	[0.11]	[0.25]	[2.12]	[2.84]
α_{FF5}	-0.19	0.12	0.04	-0.04	0.01	0.20
	[-3.39]	[2.07]	[0.71]	[-0.66]	[0.10]	[2.75]
α_{FF6}	-0.18	0.11	0.01	-0.03	0.02	0.19
	[-3.08]	[1.97]	[0.21]	[-0.44]	[0.32]	[2.66]
Panel B: Fama and French (2018) 6-factor model loadings for ACU-sorted portfolios						
β_{MKT}	1.01	0.98	0.96	1.02	0.99	-0.02
	[76.41]	[72.68]	[69.30]	[69.91]	[81.75]	[-1.11]
β_{SMB}	0.03	0.02	0.08	-0.01	0.08	0.05
	[1.53]	[1.00]	[4.00]	[-0.44]	[4.48]	[2.03]
β_{HML}	-0.10	-0.09	-0.07	-0.19	-0.08	0.02
	[-3.84]	[-3.31]	[-2.63]	[-6.84]	[-3.33]	[0.62]
β_{RMW}	0.21	-0.03	-0.04	0.02	0.23	0.02
	[8.12]	[-0.96]	[-1.29]	[0.76]	[9.54]	[0.49]
β_{CMA}	0.10	0.02	0.04	0.19	0.12	0.02
	[2.64]	[0.53]	[1.00]	[4.44]	[3.43]	[0.40]
β_{UMD}	-0.03	0.01	0.05	-0.02	-0.02	0.01
	[-1.95]	[0.55]	[3.49]	[-1.44]	[-1.56]	[0.41]
Panel C: Average number of firms (n) and market capitalization (me)						
n	453	654	783	672	459	
me ($\$10^6$)	1663	1891	1211	1778	1702	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the ACU strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015a) five-factor model, and the Fama and French (2015a) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.21	0.22	0.21	0.20	0.20	0.19
			[3.06]	[3.21]	[2.98]	[2.84]	[2.75]	[2.66]
Quintile	NYSE	EW	0.18	0.18	0.17	0.16	0.16	0.15
			[4.05]	[4.13]	[3.83]	[3.50]	[3.55]	[3.37]
Quintile	Name	VW	0.13	0.14	0.13	0.15	0.13	0.14
			[2.06]	[2.14]	[2.00]	[2.24]	[1.94]	[2.14]
Quintile	Cap	VW	0.19	0.21	0.20	0.18	0.19	0.18
			[2.62]	[3.00]	[2.74]	[2.51]	[2.56]	[2.41]
Decile	NYSE	VW	0.18	0.21	0.19	0.18	0.17	0.16
			[2.27]	[2.52]	[2.25]	[2.12]	[1.95]	[1.88]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.17	0.17	0.15	0.15	0.13	0.13
			[2.38]	[2.38]	[2.17]	[2.11]	[1.75]	[1.74]
Quintile	NYSE	EW	-0.03					
			[-0.60]					
Quintile	Name	VW	0.09	0.09	0.08	0.09	0.07	0.07
			[1.35]	[1.38]	[1.23]	[1.39]	[0.98]	[1.12]
Quintile	Cap	VW	0.14	0.16	0.14	0.13	0.12	0.12
			[1.97]	[2.21]	[1.95]	[1.84]	[1.65]	[1.59]
Decile	NYSE	VW	0.14	0.14	0.12	0.11	0.08	0.08
			[1.65]	[1.66]	[1.41]	[1.35]	[0.95]	[0.95]

Table 3: Conditional sort on size and ACU

This table presents results for conditional double sorts on size and ACU. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ACU. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ACU and short stocks with low ACU .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results

		ACU Quintiles					ACU Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.79	0.84	0.81	0.91	0.90	0.11	0.10	0.10	0.04	0.04	0.00
		[2.86]	[2.93]	[2.86]	[3.23]	[3.04]	[0.89]	[0.86]	[0.79]	[0.35]	[0.30]	[0.03]
	(2)	0.81	0.86	0.85	0.88	0.86	0.05	0.06	0.02	0.01	-0.01	-0.02
		[3.15]	[3.32]	[3.19]	[3.38]	[3.34]	[0.62]	[0.77]	[0.23]	[0.08]	[-0.13]	[-0.23]
	(3)	0.79	0.82	0.90	0.79	0.87	0.08	0.08	0.05	0.04	0.06	0.05
		[3.37]	[3.41]	[3.77]	[3.24]	[3.68]	[0.97]	[0.94]	[0.63]	[0.45]	[0.68]	[0.55]
	(4)	0.77	0.88	0.85	0.85	0.81	0.04	0.03	0.04	0.05	0.01	0.02
		[3.53]	[3.78]	[3.91]	[3.72]	[3.70]	[0.52]	[0.43]	[0.49]	[0.63]	[0.14]	[0.31]
	(5)	0.46	0.71	0.60	0.57	0.76	0.30	0.32	0.30	0.28	0.29	0.28
		[2.35]	[3.77]	[3.25]	[2.92]	[4.08]	[3.46]	[3.66]	[3.41]	[3.16]	[3.23]	[3.04]

Panel B: Portfolio average number of firms and market capitalization

		ACU Quintiles					ACU Quintiles				
		Average n					Average market capitalization (\$$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	351	350	349	349	349	34	26	27	26	33
	(2)	93	93	93	93	93	52	52	50	51	52
	(3)	64	64	64	64	64	89	87	86	88	90
	(4)	52	53	53	53	52	193	197	191	194	195
	(5)	45	45	45	45	45	1176	1494	1070	1437	1254

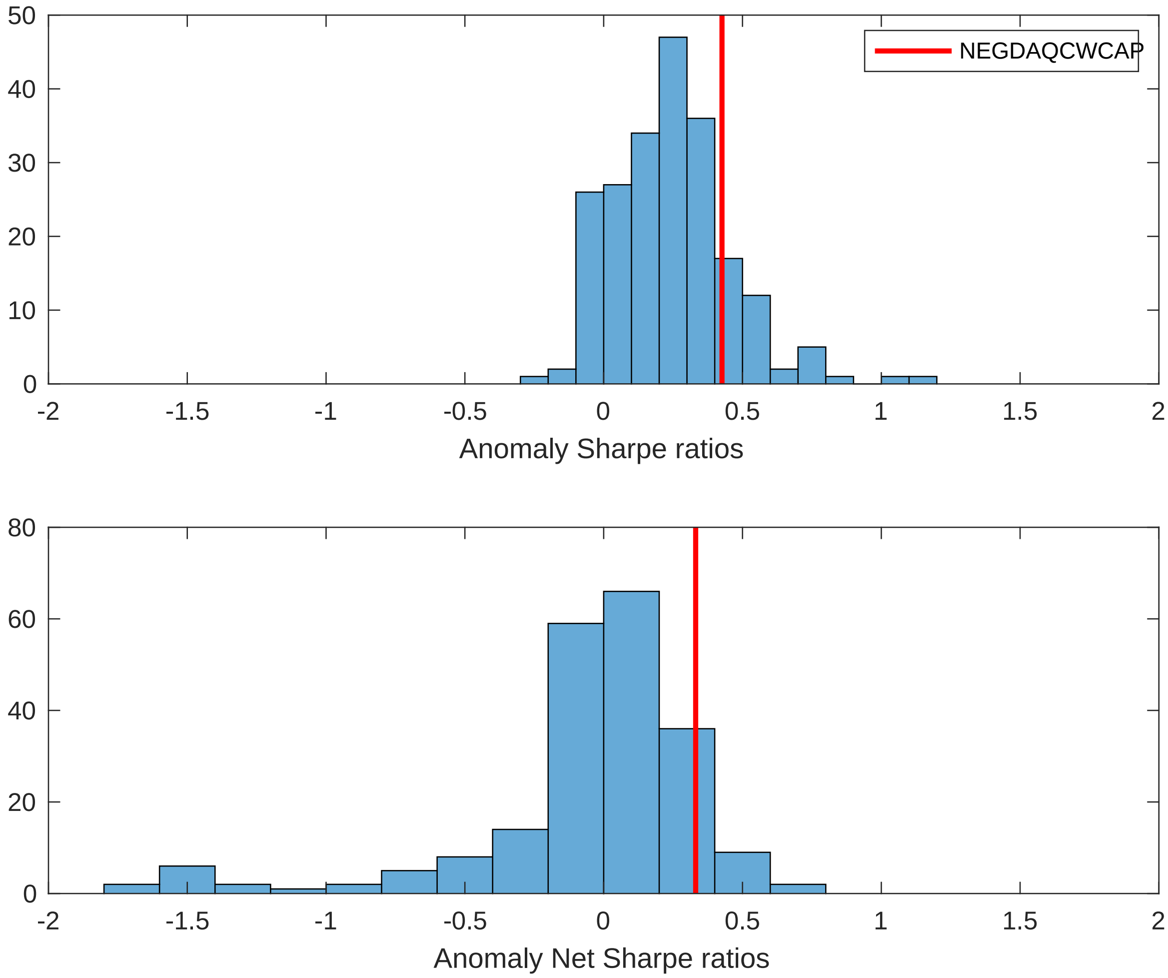


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ACU with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

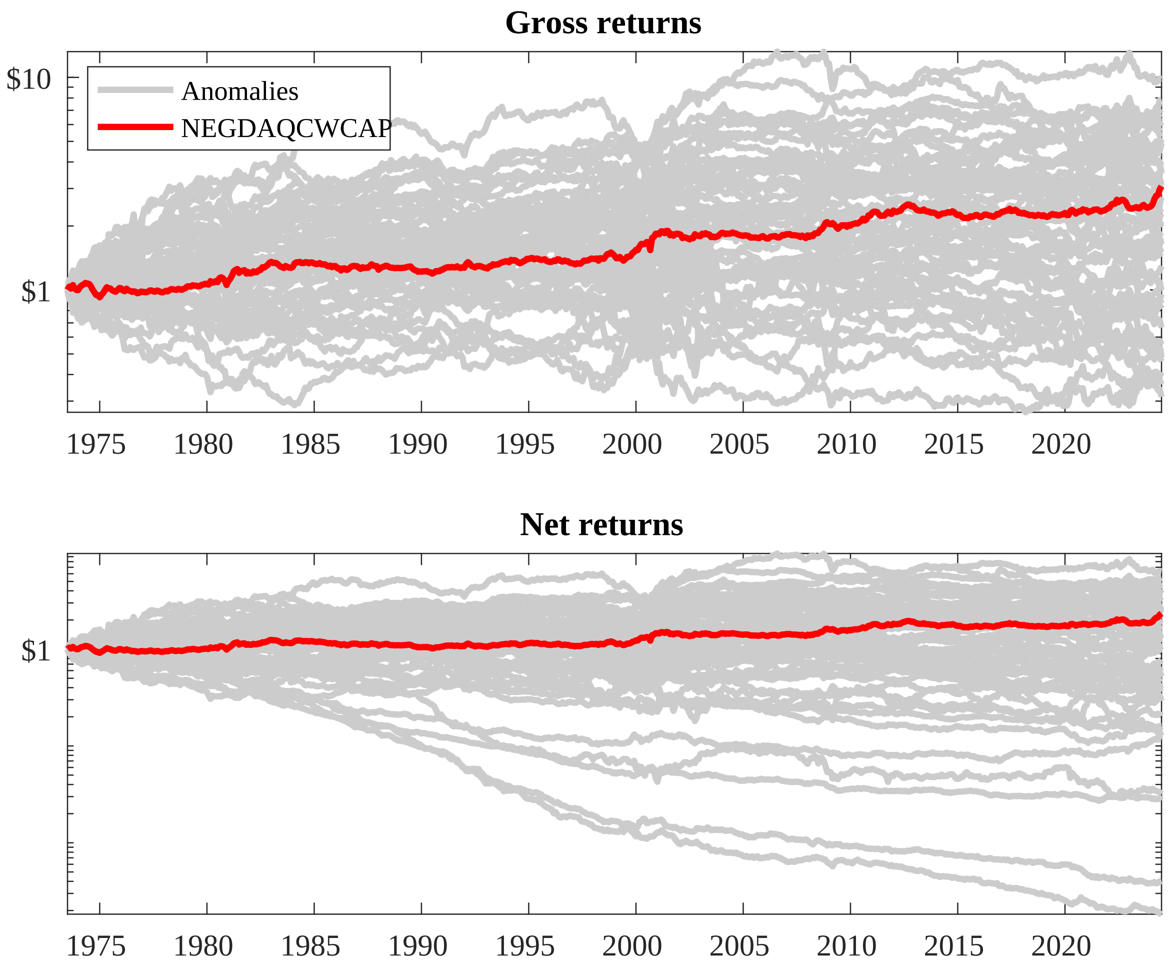
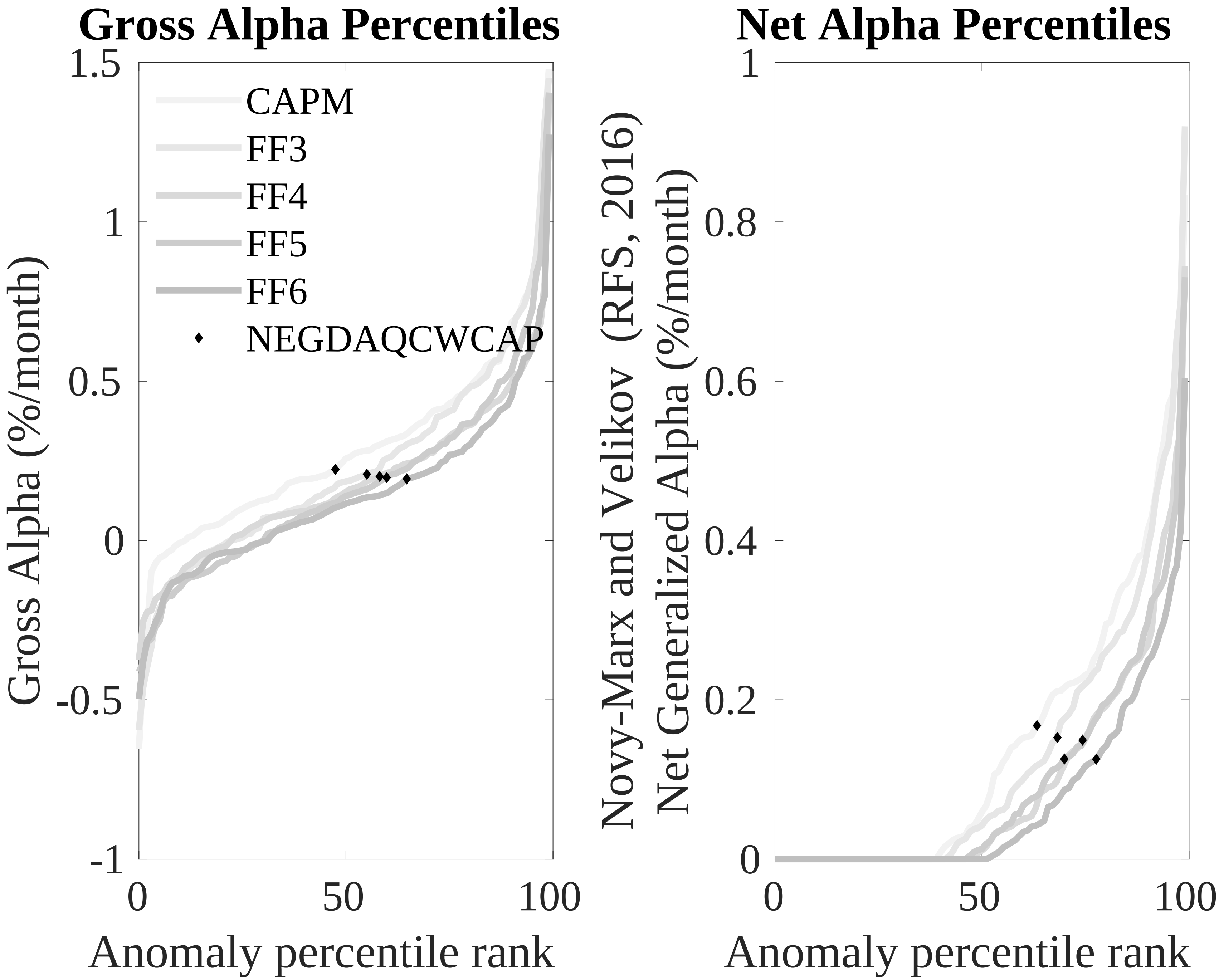


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ACU trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDAQCWCAP
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100

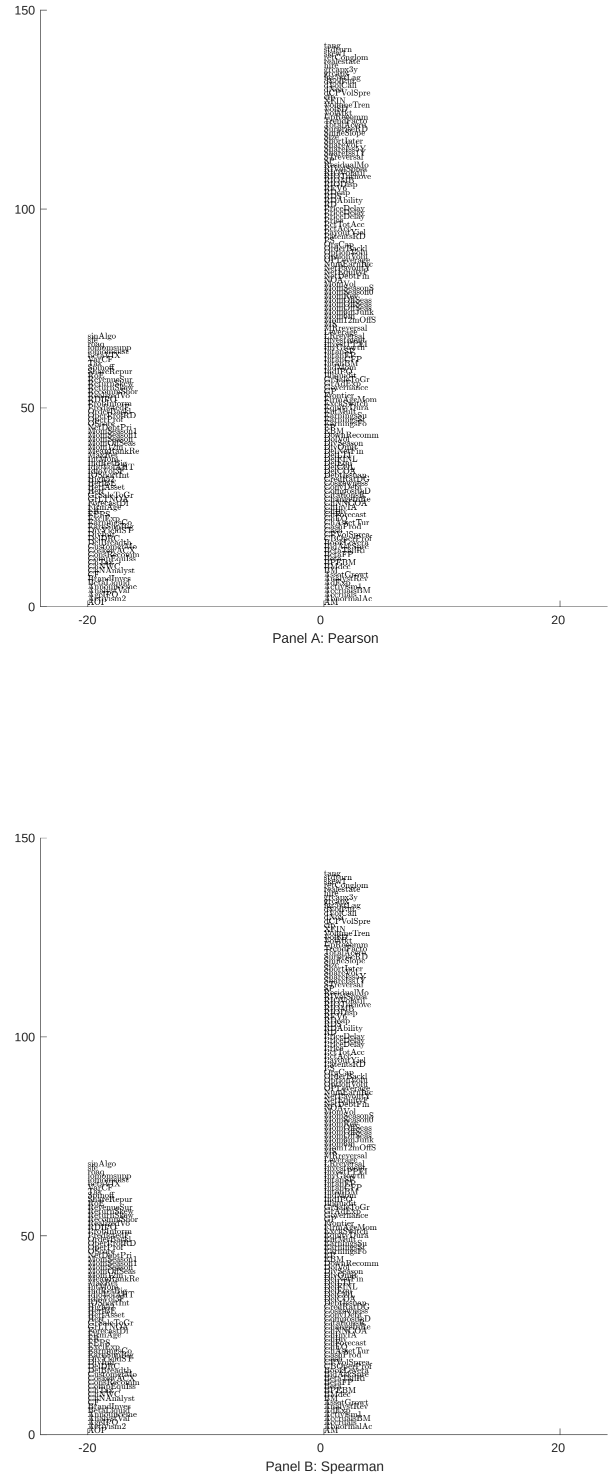


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 209 filtered anomaly signals with ACU. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

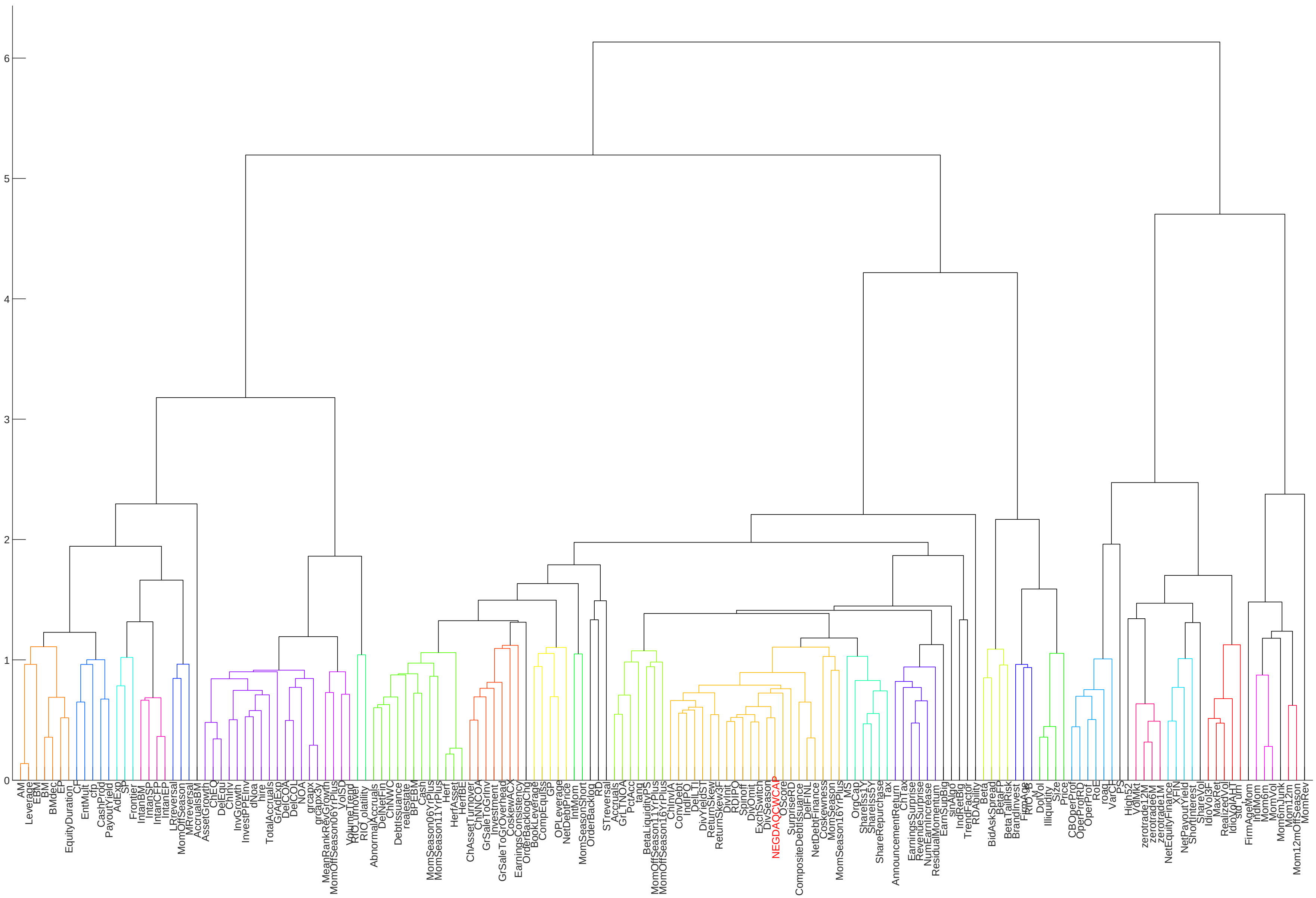


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

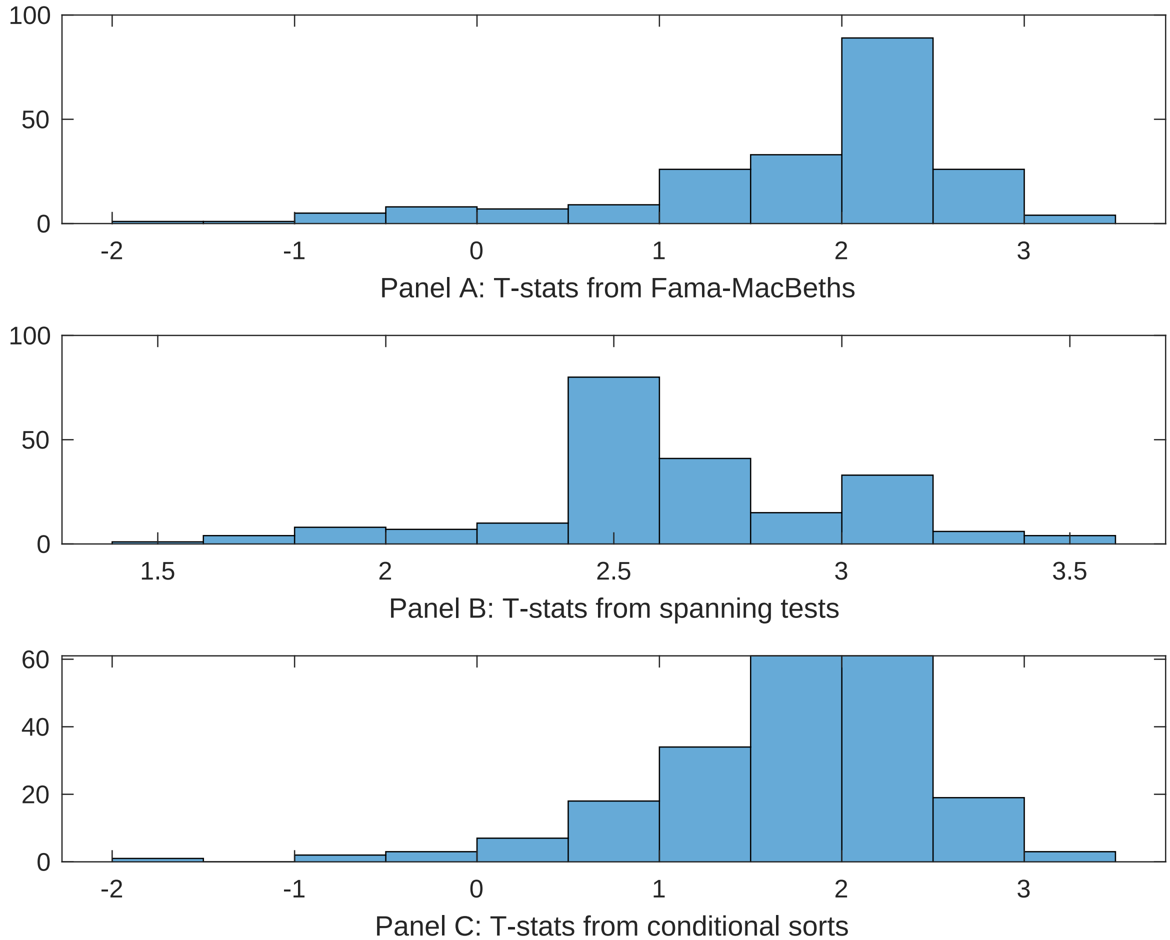


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of ACU conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ACU} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ACU} ACU_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ACU,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ACU. Stocks are finally grouped into five ACU portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ACU trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on ACU. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ACU} ACU_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Change in net financial assets, Change in Net Noncurrent Op Assets, Asset growth, Growth in book equity, Change in equity to assets, Sales growth over inventory growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13	0.13	0.14	0.19	0.14	0.13	0.15
	[5.18]	[5.09]	[5.72]	[6.95]	[5.30]	[5.18]	[6.81]
ACU	0.22	0.47	-0.17	0.52	0.49	0.53	-0.23
	[0.89]	[1.90]	[-0.65]	[2.14]	[2.01]	[2.13]	[-0.88]
Anomaly 1	0.79						0.43
	[5.24]						[1.76]
Anomaly 2		0.92					0.99
		[4.25]					[0.31]
Anomaly 3			0.11				0.84
			[8.51]				[6.06]
Anomaly 4				0.52			0.33
				[5.13]			[0.30]
Anomaly 5					0.15		0.49
					[4.35]		[0.79]
Anomaly 6						0.13	0.60
						[3.58]	[1.56]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ACU trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ACU} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015a) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Change in net financial assets, Change in Net Noncurrent Op Assets, Asset growth, Growth in book equity, Change in equity to assets, Sales growth over inventory growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.21	0.19	0.18	0.17	0.18	0.18	0.19
	[2.93]	[2.62]	[2.44]	[2.37]	[2.45]	[2.54]	[2.71]
Anomaly 1	-15.30						-5.87
	[-4.50]						[-1.46]
Anomaly 2		-15.54					-10.57
		[-4.81]					[-3.03]
Anomaly 3			13.88				3.86
			[3.31]				[0.81]
Anomaly 4				17.06			8.50
				[4.42]			[1.51]
Anomaly 5					14.49		1.86
					[4.08]		[0.35]
Anomaly 6						-10.73	-5.38
						[-3.45]	[-1.68]
mkt	-2.20	-1.95	-1.88	-1.30	-1.99	-1.68	-1.65
	[-1.33]	[-1.18]	[-1.13]	[-0.79]	[-1.21]	[-1.01]	[-1.01]
smb	4.11	5.80	3.99	4.47	5.05	5.45	4.89
	[1.64]	[2.32]	[1.57]	[1.79]	[2.02]	[2.17]	[1.95]
hml	1.89	2.22	0.90	-0.42	-0.40	1.26	0.52
	[0.59]	[0.70]	[0.28]	[-0.13]	[-0.12]	[0.39]	[0.16]
rmw	0.28	4.07	2.16	2.67	3.41	3.25	3.77
	[0.08]	[1.25]	[0.66]	[0.82]	[1.05]	[1.00]	[1.14]
cma	-0.99	2.49	-13.81	-13.35	-11.46	3.66	-13.13
	[-0.20]	[0.52]	[-1.97]	[-2.20]	[-1.91]	[0.75]	[-1.83]
umd	0.84	2.64	0.84	0.12	0.67	1.82	2.57
	[0.51]	[1.56]	[0.51]	[0.07]	[0.41]	[1.08]	[1.51]
# months	612	612	612	612	612	612	612
$\bar{R}^2$(%)	3	4	2	3	3	2	7

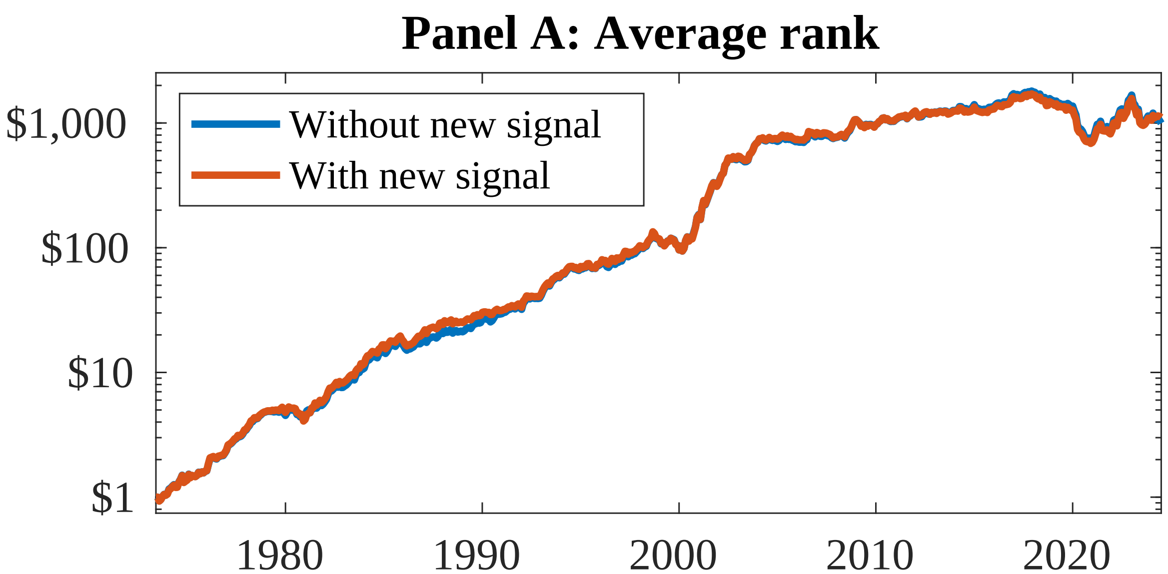


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as ACU. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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